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Ratio Analysis

How to improve your data interpretation

Session Overview

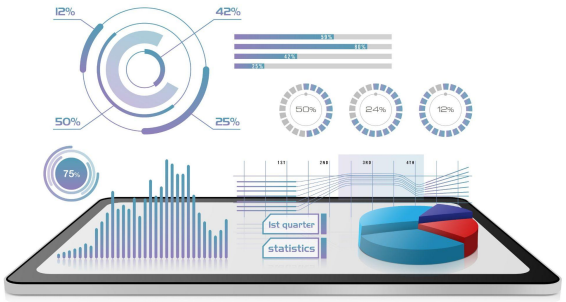
There are 5 components in business management. (We call it the 5 steps in the Advisory staircase)

1. Compilation/ Verification: (Get the data collected, ensure it has integrity and is timely)
2. Assimilation: (Render the data so it's easy to view and comprehend)
3. Interpretation: (Recognise correlation vs causation)
4. Application: (Determine the next 'experiment' - assign task ownership)
5. Observation: (Review progress, learn, adapt and go again)

The critical point about Ratio Analysis is ensuring the Assimilation is aligned with the Interpretation requirements.

You might think of it as being sure to 'Compare Apples with Apples'.

Agenda



- **Know Thyself First**
- **When data is (Less) relevant**
- **Interrogate The Benchmark Data**
- **Common Size Year on Year Data**

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Know Thyself First

Many consultants are quick to purchase benchmark data, (more accurately known as 'External Benchmark' data) without FIRST Capturing and Quantifying the INTERNAL business assumptions. This makes their financial performance data (Internal Benchmark data) non- standardised.

This is a rookie mistake.

The purpose of developing a robust predictive (revenue) model is to ensure the key business variables have been identified; AS HAS their INTERDEPENDENCE to other variables. (in english - this answers the question to 'what happens if I pull this lever'?)

Once the critical variables have been established and the operating procedures (that relate to said variables) have been defined, the monitoring and adjustment to operating procedures can begin.

The real art in management reporting lies in obtaining '3 perspectives'.

1. A predictive/scenario (revenue) model that is validated by actual performance data.
2. Clearly defined *operational objectives monitored via accurate & timely performance reports.
3. External Benchmark Data to encourage the challenge of internal beliefs.

** The operational objectives should of course, be the result of a well considered Strategic Plan; there's little point measuring your speed if you don't know where you're going.*

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When Data Is Less Relevant

There is no point comparing external benchmark ratios from within your industry, or even historical ratios for your own business if:

A: They have not been calculated in an identical manner.

You can pick up any number of different textbooks that record alternative ratio calculations; all will provide sound mathematical and business logic for their particular calculation. From a practical sense, you're looking to utilise ratios for positive change so ensure you use a ratio calculation that you understand, that requires data that you can easily generate.

For example, use credit sales instead of total sales in the debtor's days on hand ratio if this information is easily available.

DO NOT use different ratio calculations in different periods. You must be consistent with the method used to calculate a particular ratio.

If using benchmark information, ensure that you know how the ratio was calculated. DO NOT compare these with your ratios unless they have been calculated using identical methods (or make allowance for any small differences).

When Data Is Less Relevant

There is no point comparing external benchmark ratios from within your industry, or even historical ratios for your own business if:

B: One-off situations have occurred that distort the ratio for a given period.

In some cases, changes in ratios can be explained by one-off situations. As an example, a large one-off order may have been filled by your business. The debtor's days on hand figure may be a lot higher than previous years if the order has been invoiced but not paid for.

This does not demonstrate a collection issue and therefore the figure relating to the order should be removed from your debtors figure for calculation and comparison purposes.

C: The ratios belong to time periods that have fundamental differences.

Avoid direct comparison of ratios that are calculated in periods with fundamental differences. As an example, your business produces quarterly financial statements and your business is highly seasonal. During your peak season, the ratios will differ from the low season results.

This may not be due to any management action or inaction, but rather the business cycle. Either adjust for this or only compare quarters at the same stage of your seasonal cycle.

Interrogate The Benchmark Data

External Benchmarking

It can be useful to learn how your business is performing compared with others in the same industry. There are organisations that provide benchmarking data, these include:

- Universities and other tertiary institutions.
- Suppliers of business information e.g. The Hackett Group. https://www.thehackettgroup.com/ or Stats NZ https://www.stats.govt.nz/
- Business consultants

Using any of the above organisations will cost you; it is imperative therefore, that you receive data that is robust. Some of the questions to ask include:

- How many businesses does the data cover? You should be happy that the results provided are representative of your industry. This is especially concerning if there is an extreme result in a small sample.
- Can you accurately compare a category/sample to your particular business?
- Are the figures provided from businesses of similar size?
- Do the figures include upper and lower quartile figures? The further these lie from the average, the less meaningful the data is likely to be.
- How old is the data? The figures should be updated on at least an annual basis.
- Do the figures include comprehensive information on how the ratios were calculated? Will you be 'comparing apples with apples'.
- Can you compare the industry figures directly with your business? e.g. do not compare retail data with wholesale.
- Are there any other factors that may directly influence the comparison of ratios including geographical location, local labour markets, different accounting methods, quality of product, sources and methods of financing or terms of sale.
- Do the other businesses in the sample have a similar seasonal cycle to yours? A business exporting may provide the same product at a different time of year. This may influence the ratio at peak and low season.

Common Size Year on Year Data

Imagine you work as a 'crew-chief' on a drag race car (rail). You're attempting to improve your car and make changes between each race throughout the day. However, in this situation, your car starts ahead or behind your competitor based on how you performed **the race before.** How can you determine if you're doing better or worse if the ONLY measure of success is based on if you cross the line first?

Assessing your business can be similar if you compare year on year data without adjusting for the previous year's performance.

As an example, you may have had a great previous year so begin the next with more cash reserves and increased stock. The additional cash means you can buy greater stock outside of normal purchasing cycles and you can run down past stock levels whilst your competitors are paying peak prices. If you compare the second year (to the first) it might appear you're doing 'even better' and it's possible you may confuse other management decisions as being causal to your success. Remember, you started with an advantage.

To avoid confusion and confirmation bias, you need to adjust your data to measure BOTH your 'Start' position AND your 'Finish' position, for each year.

Common Size Balance Sheet

Divide each item in the balance sheet by **Total Assets.**

Common Size Profit & Loss Account

Divide each item in the balance sheet by **Total Sales.**

By comparing the common size items, the business can compare all items against previous years.

Theory and Core Philosophy

Effective ratio analysis is more than calculating financial metrics; it is a collaborative process designed to significantly improve data interpretation. The core philosophy revolves around the principle of ensuring you always **"Compare Apples with Apples"**.
Traditionally, accountants and consultants compile reports and simply hand them to clients.

However, this framework advocates for a fundamental shift: instead of just delivering a report, you must create a shared, interactive space where the advisor and the business owner analyze the findings and draw conclusions together. Getting the client heavily involved ensures they become educated, engaged, and motivated to use the data for future decision-making.

Providing the client with raw data without requiring them to put effort into understanding it means they are unlikely to assign it any meaning or turn it into new actions.

By creating this collaborative environment, consultants do not lose value or status; rather, the relationship grows as both parties identify key insights together.

Step By Step

Implementing this methodology relies on navigating the **5 Steps in the Advisory Staircase:**
1.Compilation, 2. Assimilation, 3. Interpretation, 4. Application, and 5. Observation.

Follow these steps to put the theory into practice:

Step 1: Know Thyself First (Internal Modeling) Before looking outward, you must capture and quantify your own internal business assumptions to ensure your financial performance data is standardized.

- Build a predictive revenue model: Identify your key business variables and determine how they are interdependent (e.g., answering "what happens if I pull this lever?").
- Validate the model: Run your "perfect world" model, compare it against actual performance data, and investigate any major discrepancies until the model accurately reflects reality.
- Define operational objectives: Ensure your objectives align with a considered Strategic Plan, because there is no point in measuring your speed if you do not know where you are going.

Step 2: Collect and Assimilate the Data

- Compilation: Gather the data, ensuring it is timely and maintains high integrity.
- Assimilation: Render the data so it is easy to view. At this stage, it is highly recommended to use manual tools like Google Sheets rather than fully automated fancy software. Having the client physically cut, paste, and input the data makes the process tactile; just as a baby must chew steak to digest it, a client must interact with the data to truly absorb and interpret it.

Step 3: Clean and Standardize (Common Sizing)

To properly interpret the data, you must adjust for any inherent advantages or disadvantages carried over from previous periods. Think of business performance like a drag race where your starting position is determined by how well you did the year before; to measure true success, you must adjust the data to evaluate both the "start" and "finish" positions for each specific year.

- Create a Common Size Balance Sheet: Divide every item on the balance sheet by Total Assets.
- Create a Common Size Profit & Loss Account: Divide every item in the P&L by Total Sales.
- Filter out irrelevant data: Ensure you remove data distorted by fundamental differences (like peak vs. low seasonal cycles), calculation inconsistencies, or major one-off events (such as a massive singular order that falsely inflates debtor days).

Step 4: Interpret the Findings Internally

- Interpretation: Work together with the client to recognize true causation versus mere correlation in the data. Ensure that the way you assimilated the data perfectly aligns with these interpretation requirements.

Step 5: Introduce External Benchmarks

A common "rookie mistake" is to purchase external benchmark data before understanding internal assumptions. Only bring in benchmark data after your internal model is solidified.

- **Challenge your beliefs:** Use external benchmark data to see if your approach is myopic and to challenge your internal beliefs, not as a pass/fail yardstick to judge your business against others.
- **Interrogate the benchmark:** Always heavily question the purchased data. Ask if the sample size is representative, if the businesses are of similar size, if the calculation methods are identical to yours, and if geographical or seasonal factors might skew the results.

Step 6: Apply and Observe

- **Application:** Based on your shared interpretation, determine the next "experiment" for the business and assign clear task ownership.
- **Observation:** Review the progress of these new actions, learn from the outcomes, adapt your operating procedures, and repeat the cycle.

Common Q&A

Q: Why shouldn't I immediately compare my business to external industry benchmarks?
A: Comparing against external benchmarks prematurely is a rookie mistake. If you haven't first modeled your own business and quantified your internal assumptions, your data will not be standardized. External benchmarks should be used to challenge your internal beliefs to ensure you aren't being myopic, rather than being used as a simple pass/fail grade.

Q: Why do you suggest using basic spreadsheets where clients have to manually input data instead of automated reporting software?
A: Manual data entry forces the client to be involved in a tactile way. If you just hand them a finished, automated report, they are unlikely to make the effort to assimilate the information. By investing a little bit of time to interact with the data, they mentally "digest" it, which allows them to assign meaning to it and ultimately use it to take new actions.

Q: In what situations is historical or benchmark data completely irrelevant?
A: Data is useless for comparison if the ratios were calculated using different mathematical methods. Furthermore, you should avoid comparisons if the data includes large, one-off situations that distort the period, or if you are comparing periods with fundamental business cycle differences, such as comparing a peak season to a low season.

Q: What is "Common Size Year on Year Data" and why is it necessary?
A: Common sizing involves converting financial figures into percentages (e.g., dividing all balance sheet items by Total Assets) so that you can fairly compare items against previous years. It is necessary to eliminate confirmation bias and prevent business owners from confusing an inherited starting advantage (like leftover cash reserves from a highly successful prior year) with current management success.